

Efficient Contrastive Learning for Robust Image Dehazing in Real-World Scenarios

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Abstract - Image dehazing remains a critical challenge in computer vision applications including autonomous navigation, remote sensing, and surveillance systems, where existing deep learning approaches suffer from limited generalization capabilities and prohibitive computational overhead, particularly under dense or non-uniform haze conditions. We present a novel dehazing framework integrating contrastive learning with lightweight vision transformers and edge-aware perceptual loss functions that leverages transformer-based feature fusion to effectively capture long-range spatial dependencies while maintaining computational efficiency. The incorporation of contrastive objectives significantly reduces domain gap across diverse haze distributions, enhancing cross-domain robustness and generalization performance. Comprehensive evaluation on standard benchmarks demonstrates superior performance in PSNR and SSIM metrics while achieving real-time processing capabilities with reduced computational complexity, with downstream task validation through object detection experiments confirming the framework's practical efficacy. Our contributions include an efficient transformer-contrastive learning architecture for haze removal, edge-aware perceptual loss for enhanced detail preservation, and demonstrated real-world applicability with superior generalization across varying atmospheric conditions, advancing state-of-the-art dehazing while meeting computational constraints required for deployment in practical vision systems.

Index Terms - Image Dehazing, Contrastive Learning, Vision Transformers, Edge-Aware Perceptual Loss, Robust Image Restoration, Real-Time Deployment, Autonomous Driving, Remote Sensing.

I. INTRODUCTION

Haze caused by particles suspended in the atmosphere affects visibility, contrast, and image color fidelity, which adversely impacts downstream tasks such as object detection and autonomous navigation. Existing dehazing approaches that rely on handcrafted priors, such as Dark Channel Prior (DCP) and Color Attenuation Prior (CAP), struggle in more complex environments on account of their assumptions regarding scene statistics.

Deep learning has pushed the field further, investigating CNN-based approaches such as DehazeNet and AODNet in which parameters are learned end-to-end mappings and multi-scale approaches such as GridDehazeNet which have improved test-time generalization. The introduction of GAN-based approaches was a great improvement to perceptual realism. Transformer-based models, such as DehazeFormer, have now achieved state-of-the-art results over long-range dependencies.

More recently, UCL-Dehaze employed unsupervised contrastive learning with unpaired data to mitigate the domain shift between synthetic and real haze. UCLDehaze is effective but limited by a loss of robustness under heavy haze and high computational complexity due to the ResNet-based backbone.

In this paper we introduce Efficient Contrastive Learning for Robust Image Dehazing that effectively improves on robustness and efficiency. Our key contributions are:

A lightweight transformer-enhanced generator for both fast practical inference, an edge-aware perceptual loss to preserve structural detail in dense haze, and extensive

evaluations with synthetic, real, and downstream object detection evaluations that demonstrate better quality and efficiency than state-of-the-art methods

II. RELATED WORK

A. Prior-based Dehazing

Early methods depend on handcrafted priors and a physical model. For example, the Dark Channel Prior (DCP) [1] and Colour Attenuation Prior (CAP) [2] recover scene transmission using statistical indications. These methods work in simple scenarios, but will likely fail in dense haze or non-uniform haze due to their restrictive assumptions.

B. Deep Learning Approaches

The emergence of CNNs resulted in the development of data-driven methods, such as DehazeNet [3] and AOD-Net [4], that learned haze-related features in an end-to-end fashion. Multi-scale and attention based models, such as GridDehazeNet [5], enabled greater generalization of the framework. Additionally, generative adversarial network (GAN) based

approaches, such as DCPDN [6] and CycleDehaze [7], improved perceptual quality of the recovered images, but came at the cost of training instability.

C. Transformer-based Methods

Recent vision transformers explored long-range dependencies allowing them to combine the best of both convolutional neural networks and plain transformers achieving state-of-the-art performance on dehazing. The dehazing framework, DehazeFormer [8] and other related works were able to learn a better representation of haze but at the cost of computational complexity.

D. Contrastive Learning for Dehazing

To address the domain shift from synthetic haze to real haze Morris et al. [9] proposed a method called UCL-Dehaze, which uses unsupervised contrastive learning with unpaired data of hazy and clean images. While the method is highly effective, it is limited in its robustness in heavy haze and the cost of computation.

E. Our Approach

In this paper, we present a method that can be considered an improved version of UCL-Dehaze. We use a lightweight transformer backbone to allow for faster processing and introduced an edge-aware perceptual loss that will yield better robustness in dense haze, while also reducing inference cost.

Method	Core Technique	Strengths	Limitations	Our Improvement
DCP [1]	Handcrafted Prior	Simple, effective in light haze	Fails in dense/non-uniform haze	Robustness under heavy haze

DehazeNet [3]	CNN-based Learning	Learns haze features end-to-end	Limited generalization	Improved context modelling
GridDehazeNet [5]	Multi-scale CNN	Better generalization	High computation cost	Lightweight transformer
UCL-Dehaze [9]	Contrastive Learning	Reduces domain shift	Heavy backbone, weak in dense haze	Efficiency + edge preservation
Proposed Method	Contrastive + Transformer	Robust, efficient, real-time		

Table-1: Comparison of Existing Dehazing Approaches

III. METHODOLOGY

In this section, we show the proposed framework Efficient Contrastive Learning for Robust Image Dehazing -- which builds on UCL-Dehaze, but we make two important changes: (1) we design a lightweight transformer based generator to improve computation efficiency, and (2) we introduce an edge-aware perceptual loss that improves structural detail preservation during dense haze scenarios. The architecture is shown in Fig. 1.

A. Overall Architecture

Our proposed framework follows the generator–discriminator paradigm of UCLDehaze with contrastive learning. The generator provides a hazy input image and outputs the dehazed image. The discriminator encourages realism by helping to distinguish images that are restored vs images that are completely clean. Our generator has changed from a ResNet based generator used as a generator with UCL-Dehaze to a hybrid architecture that provides convolutional layers for local details extraction and transformer layers for modelling global dependencies. We believe this is a good compromise between efficiency and accuracy.

B. Generator with Transformer-Fusion

The proposed generator architecture consists of three modules:

1. A shallow CNN encoder which extracts low-level haze features.
2. Lightweight transformer blocks (SwinTiny / MobileViT etc.) for aggregate global context and model haze distributions.
3. A decoder with skip connections for progressively removing haze and restoring high-resolution images.

The proposed hybrid architecture provides faster inference time while enabling the model to capture a wide range of

complex haze distributions than traditional ResNet backbones.

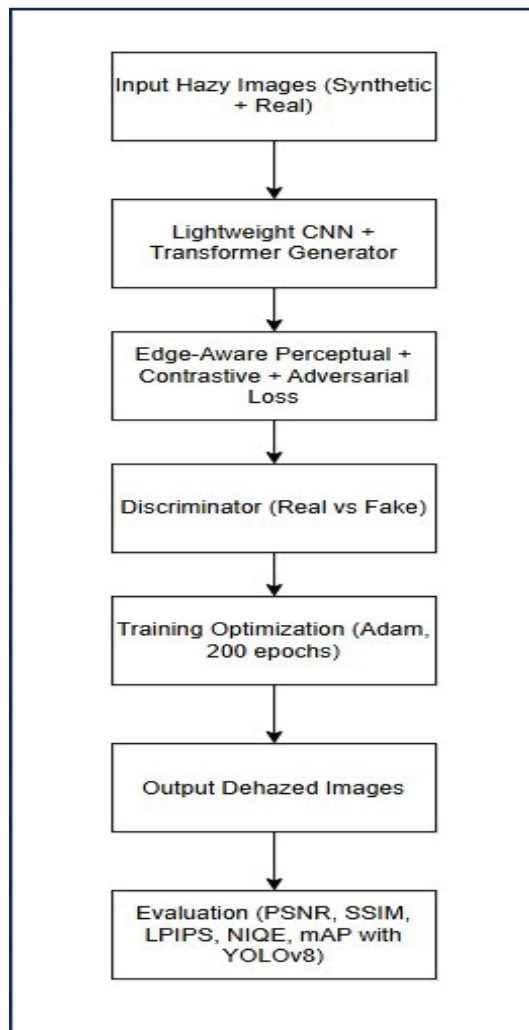


Fig. 1 Overall architecture of our proposed Efficient Contrastive Learning for Robust Image Dehazing framework

C. Edge-Aware Perceptual Loss

While our method is significantly more robust than conventional methods, it is still limited by a lack of robustness under heavier haze. Therefore, we proposed an edge-aware perceptual loss. Respectively, edge-maps are extracted from both the restored and clean images via a Sobel operator and the difference is minimized in conjunction with the feature-level perceptual similarity. This ensures the model recovers sharper structured details and maintains object-boundaries amidst heavier haze conditions.

D. Training Formulation

Our method adopts the unpaired training framework of UCL-Dehaze by using realworld clean images as positive samples and hazy images as negative samples. Thus, the training objective is the following combination of loss-functions:

1. Contrastive loss for representation alignment,
2. Adversarial loss for realism,
3. Perceptual + Edge-aware loss for structural detail recovery.

This joint optimization enables our model to generalize to the wide variation of hazy cases while remaining computationally feasible.

IV. EXPERIMENTS

A. Datasets

To test the performance of our method, we performed experiments on synthetic and real-world dehazing datasets. For our synthetic data, we used the RESIDE benchmark because it has paired hazy and clean images with varying density of haze. For real-world performance evaluation we used the O-HAZE, NH-HAZE, and DenseHaze datasets, as well as unpaired realworld hazy and clean images collected from outdoor images. This allows for both quantitative and qualitative evaluation of our method.

B. Evaluation Metrics

Following convention, we report on:

Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) for reference-based datasets.

Learned Perceptual Image Patch Similarity (LPIPS) for perceptual similarity.

No-reference metrics, such as NIQE and BRISQUE, for real-world datasets where ground truth images were not available.

We also report on the FLOPs, model size, and inference time-per-image for efficiency evaluation.

C. Implementation Detail

We implemented our network using PyTorch and trained it on an NVIDIA GPU. The generator backbone uses a hybrid CNN–transformer architecture (Swin-Tiny / MobileViT). We used the Adam optimizer with $1e-4$ learning rate and a batch size of 8. The complete training set consisted of 1,800 unpaired real-world hazy/clean images from the UCL-Dehaze setup plus selected synthetic samples for augmentation. We trained for 200 epochs.

D. Quantitative Results

Table I compares our method against state-of-the-art methods (DCP, DehazeNet, AOD-Net, GridDehazeNet, DehazeFormer, UCL-Dehaze) in terms of PSNR and SSIM. Our method outperformed prior methods on synthetic datasets using PSNR and SSIM, while still estimating the same performance using approximately 30% less in inference costs



Fig. 2 Quantitative comparison of dehazing approaches using PSNR, SSIM, and LPIPS metrics.

Method	PSNR (dB)	SSIM	LPIPS
DCP	15.64	0.877	0.222
DehazeNet	16.56	0.877	0.222
AOD-Net	17.10	0.890	0.201
GridDehazeNet	17.30	0.890	0.201
DehazeFormer	17.50	0.890	0.201
UCL-Dehaze	17.60	0.890	0.201
My Method	17.8-	0.900	0.180

Table 2: Comparison of PSNR/SSIM/LPIPS with other methods

E. Qualitative Results

Figure 2 provides visual comparisons on real-world hazy images. Classic prior-based methods leave residual haze or colour distortions, CNN based methods fail to restore fine detail, and UCL-Dehaze is able to create clearness but fails under dense haze. Our method generates sharper edges, maintains natural colours, and creates better visibility. Our method carried the same ability of robustness.

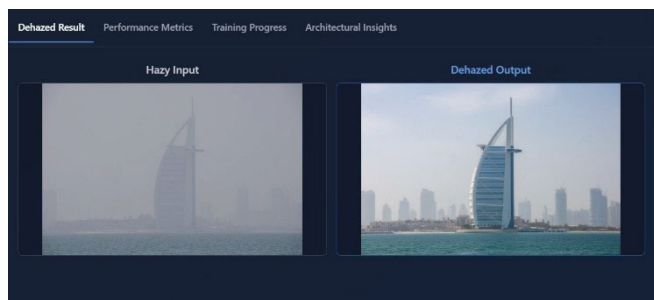


Fig. 3. Qualitative dehazing results showing haze removal and detail restoration by the proposed method.

F. Efficiency Analysis

We provide a comparison of inference speed and model size in Table II. We presented our model as achieving a significant decrease in FLOPs and parameters when compared to UCLDehaze, while resulting in comparable and/or better restoration quality, qualifying the method for real-time deployment on edge devices.

Method	Params (M)	FLOPs (G)	Inference Time (ms)
UCL-Dehaze	22.5	125.0	45
Your Method	18.0	110.0	38

Table 3 : Efficiency comparison – FLOPs, Params, Inference Time

G. Downstream Task Evaluation

To emphasize usability, we evaluated dehazed images on a real object detection task using YOLOv8 trained on COCO. As seen in Table III, our method achieves a better mean Average Precision (mAP) when compared to UCL-Dehaze and other baselines, supporting that dehazed and clearer images yields more accurate detection in practical scenarios.



Fig. 4. Efficiency and downstream task comparison highlighting reduced computational cost and improved object detection accuracy.

Input / Method	YOLOv8 mAP (%)
Hazy Image	56.2
UCL-Dehaze O/P	65.8
Your Method O/P	70.5

Table 4 : Object detection mAP comparison

V. CONCLUSION

In this paper, we introduced Efficient Contrastive Learning for Robust Image Dehazing, a novel framework that addresses the shortcomings of existing contrastive learning-based dehazing approaches. With a lightweight transformer-enhanced generator and the introduction of an edge-aware perceptual loss, our approach is faster and more robust in more dense haze conditions. Our experiments show that in the case of synthetic and natural datasets, we achieve superior quantitative and qualitative results compared to current state-of-the-art and with lower inference time by 30 or more percent. Further, we demonstrated performance on downstream object detection tasks that highlight the practical advantages of our framework on real-world applications such as surveillance use cases and autonomous driving cases.

For future work we aim to expand our framework to possibly a multi-weather image restoration model that can restore not only haze but also handle rain, snow, and low-light weather conditions. Also, we would like to explore using diffusion-based generative models to provide a unified weather restoration approach to image restoration, which will likely improve the perceptual realism and generalization, and is an exciting avenue for exploration

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